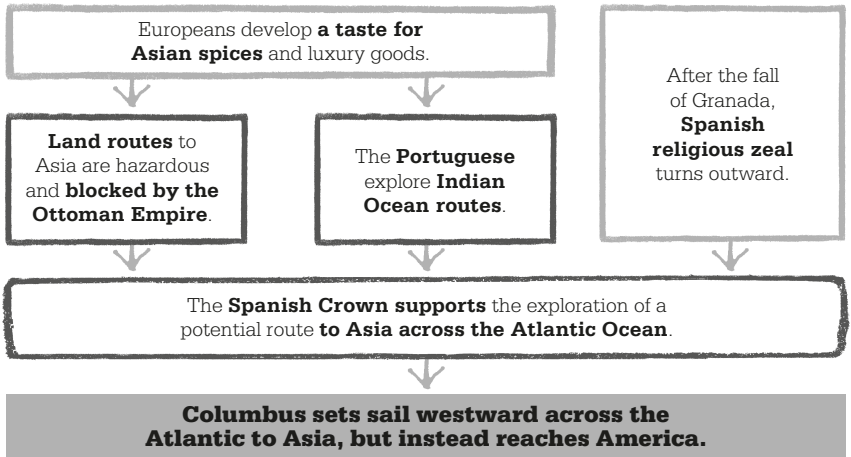




reasons to develop sea routes: anyone who could find a more direct way of importing these goods to Western Europe would become very rich.

Another reason why Europeans started to explore sea routes in the late Middle Ages was to investigate the possibility of establishing European colonies in Asia. These could act not only as trading posts, but also as bases for missionaries, who could convert the locals to Christianity. This they believed would help reduce the perceived threat of Islam.

By the 14th and 15th centuries, the Spanish, Portuguese, English, and Dutch had developed ocean-going ships, and trained sailors who could navigate over long distances. Explorers used various types of vessels, among the most successful of which was the caravel—a fast, lightweight, and extremely maneuverable ship that was usually equipped with a mix of square and lateen (triangular) sails. The lateen sails made it possible to sail to windward (into the wind), which allowed explorers to make progress even in variable wind conditions. Explorers also used the carrack, or nau, a larger vessel that was similarly rigged. On his first transatlantic voyage, Columbus took two caravels, each probably of 50–70 tons, and one carrack of about 100 tons, the extra capacity being useful for carrying stores.

Skills and technology quickly developed in both shipbuilding and navigation. Sailors used the cross-staff—a basic sighting device—or later a mariner's astrolabe, to calculate a vessel's latitude. They achieved this by measuring angles, such as the angle of the sun to the horizon. They used a magnetic compass to gauge direction, and their charts and knowledge of prevailing winds and currents improved with each voyage.

Portuguese navigators

European navigators had been striking out into the Atlantic for many decades. Sailors from Bristol, England, were sailing in the 1470s in search of a mythical island called “Brasil,” thought to be west of Ireland. The Portuguese established trading colonies on Madeira, and Prince Henry the Navigator, son of Portugal's King John I, commissioned numerous journeys of exploration to the Azores in the 15th century. Henry had started the first school for oceanic navigation, with an astronomical observatory at Sagres, Portugal, in about 1418. Here he promoted the study of navigation, map-making, and science. Henry sent ships down the west coast of Africa, to which he was particularly attracted by the potential to trade in enslaved people and gold. His ships pushed southward, setting up trading posts along the way. Subsequent rulers continued to sponsor voyages and, in 1488, Portuguese »